Precalculus Review for Calculus

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Functions and composition**

Let $f(x) = \sqrt{x - 1}$ and $g(x) = x^2 + 3$. Find:

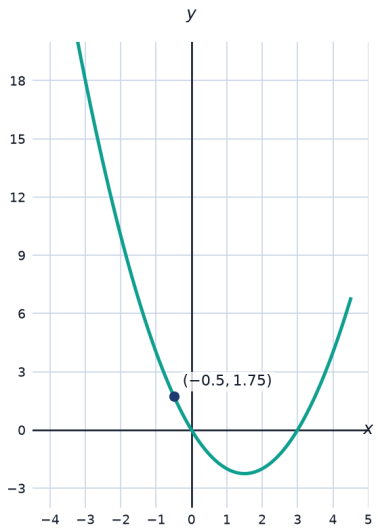
a $(f \circ g)(x)$

b $(g \circ f)(x)$

c the domain of $f \circ g$

2 **[Foundational]** • **Target: Functions and composition**

Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = x^2 - 3x$.



3 **[Foundational]** • **Target: Lines, exponentials, and trig — the toolkit calculus builds on**

Find the slope and equation of the line through $(-1, 4)$ and $(3, -2)$.

4 **[Foundational]** • **Target: Lines, exponentials, and trig — the toolkit calculus builds on**

Solve $2e^{3x} = 14$ exactly and to three decimal places.

5 **[Foundational]** • **Target: Lines, exponentials, and trig — the toolkit calculus builds on**

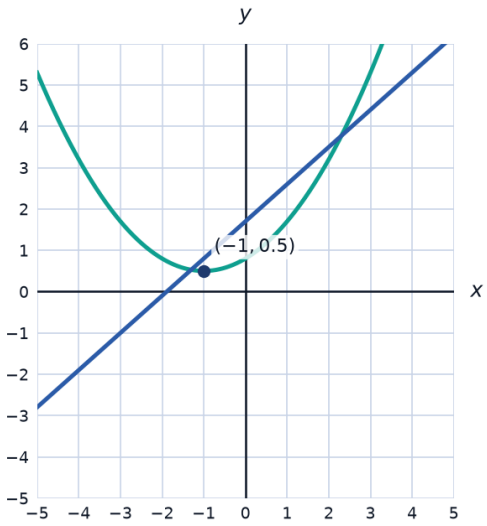
State the exact value of each expression, needed frequently when differentiating trig functions:

a $\sin\frac{\pi}{4}$ **b** $\cos\frac{\pi}{3}$ **c** $\tan\frac{\pi}{6}$

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6 [Foundational] • Target: Factoring and rational expressions

Factor each expression fully:



a $x^2 - 5x + 6$ **b** $4x^2 - 9$ **c** $x^3 - 8$

7 [Foundational] • Target: Factoring and rational expressions

Simplify $\frac{x^2 - 4}{x^2 - x - 6}$, stating the value(s) of x excluded from the domain.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

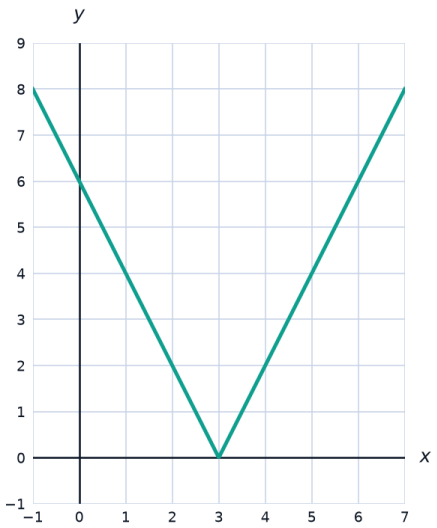
8 [Intermediate] • Target: Inverse functions and logarithms

Find the inverse of $f(x) = 3x - 5$, and verify by computing $f(f^{-1}(x))$.

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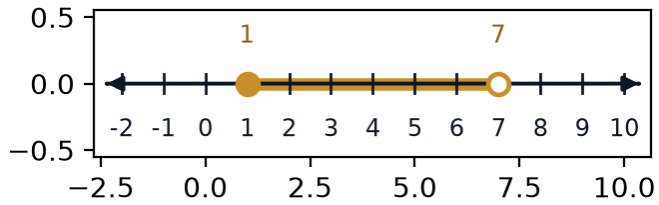
9 [Intermediate] • Target: Piecewise and absolute value review

Sketch-reasoning: rewrite $f(x) = |2x - 6|$ as a piecewise function without absolute value bars.



10 [Intermediate] • Target: Piecewise and absolute value review

Solve the inequality $|3x + 1| \leq 7$, writing your answer in interval notation.

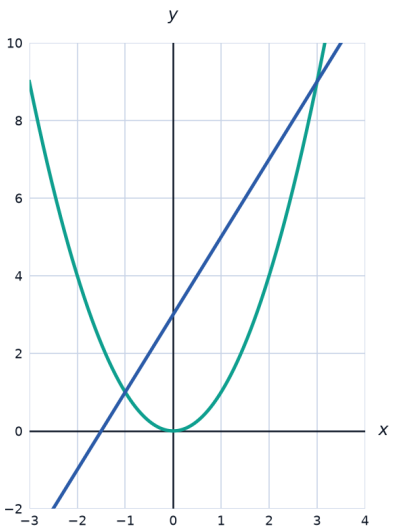


11 [Intermediate] • Target: Systems of equations

Solve the system $2x + y = 7$ and $x - y = 2$ by elimination.

12 [Intermediate] • Target: Systems of equations

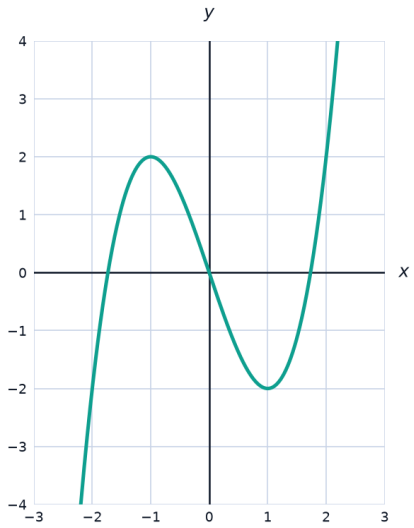
Find the point(s) of intersection of $y = x^2$ and $y = 2x + 3$.



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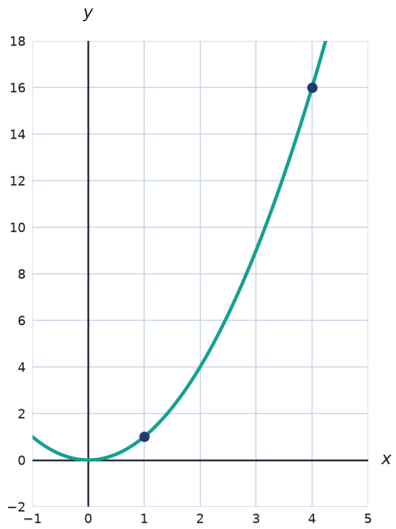
13 [Intermediate] • Target: Graphing functions calculus will build on

The graph of $f(x) = x^3 - 3x$ is shown. Use it to estimate the x-intercepts and the local max/min locations — features calculus will let you find exactly.



14 [Intermediate] • Target: Average rate of change (a preview of the derivative)

Find the average rate of change of $f(x) = x^2$ over $[1, 4]$, and interpret this as the slope of a secant line.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Average rate of change (a preview of the derivative) • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

For $f(x) = x^2$, find the average rate of change over $[1, 1 + h]$ in terms of h, and describe what happens as $h \to 0$ (a preview of the derivative).

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16 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 6.01 Related Rates

Simplify each expression, writing your answer with positive exponents:

a $x^{1/2} \cdot x^{1/3}$ **b** $(x^{2/3})^3$ **c** $\frac{x^{5/2}}{x^{1/2}}$

17 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 6.02 Optimization and Curve Sketching

Rewrite $\sqrt[3]{x^2}$ using rational exponent notation, and evaluate it at $x = 8$.

18 [Challenging] • Target: Interval notation and set operations review • Cross-Lesson: 7.01 Integration by Substitution

Write the solution set of $x^2 - 4 > 0$ in interval notation.

19 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 7.02 Integration by Parts

Level 1 · FoundationsFind the exact natural domain of the function $g(x) = \sqrt{\frac{x-1}{x+3}} - \ln(5 - x)$ in interval notation.

20 [Challenging] • Target: Interval notation and set operations review • Cross-Lesson: 8.01 Area Between Curves

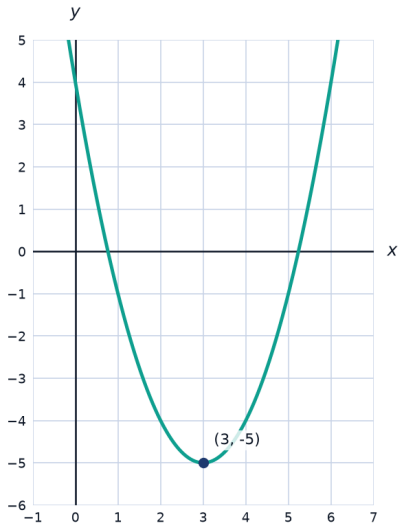
Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Functions and composition

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown. The vertex of the parabola is at the point $(3, -5)$. Find the value of the parameter k.



a. -5

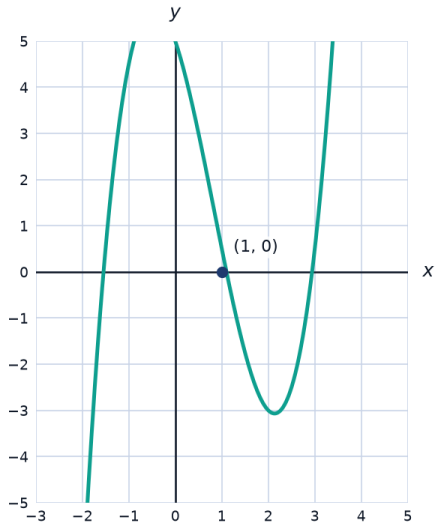
b. 4

c. 9

d. 0

22 [Foundational] • Target: Lines, exponentials, and trig — the toolkit calculus builds on

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?



a. $(-2, 3)$

b. $(0, -1)$

c. $(1, 0)$

d. $(3, 4)$

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23 [Foundational] • Target: Factoring and rational expressions

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:

a. $|\mathbf{b} - \mathbf{a}|$

b. $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$

c. $|\mathbf{d}_1 - \mathbf{d}_2|$

d. 0

24 [Foundational] • Target: Inverse functions and logarithms

Let $F(x) = \int_2^x (t^2 - 1)dt$. According to the Fundamental Theorem of Calculus, what is the derivative of $F(x)$ with respect to x?

a. $x^2 - 1$

b. x^2

c. $x^2 - 1 - (2^2 - 1)$

d. $2x$

25 [Foundational] • Target: Piecewise and absolute value review

What is the maximum rate of change of $f(x, y) = x\cos(y) - 3y$ at $(2,\ 3)$?

a. $3\sin^2(3) + 12\sin(3) + 10$

b. $2\sin(3) - \cos(3) + 3$

c. $\sqrt{3\sin^2(3) + 12\sin(3) + 10}$

d. $2\sin(3) + 3$

26 [Foundational] • Target: Systems of equations

Consider the function $f(x) = 2x^2 - 3x + 1$. What is the value of $f(3)$?

a. 10

b. 13

c. 16

d. 19

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27 [Foundational] • Target: Graphing functions calculus will build on

A spherical balloon is inflated at 4 cm³/s. Rate of radius change when $r = 2$:

a. $\frac{1}{\pi}$

b. $\frac{1}{2\pi}$

c. $\frac{1}{4\pi}$

d. $\frac{1}{8\pi}$

28 [Foundational] • Target: Average rate of change (a preview of the derivative)

Symmetric equations of line through (x_0, y_0, z_0) with direction (a, b, c):

a. $ax + by + cz = 0$

b. $x/a = y/b = z/c$

c. $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$

d. $x + y + z = 1$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Radicals and rational exponents

Net change in population from $t = 0$ to $t = T$ with rate $r(t)$:

a. $r(T) - r(0)$

b. $\int_0^T r(t)\,dt$

c. $r'(T)$

d. $r(0) \cdot T$

30 [Intermediate] • Target: Interval notation and set operations review

A 200-L tank holds brine with 20 kg of salt. Brine with $\frac{1}{2}$ kg/L enters at 10 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 20 min?

a. $100 - \frac{80}{e}$

b. $100 - \frac{100}{e}$

c. 120

d. $\frac{20}{e} + 100$

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31 [Intermediate] • Target: Functions and composition

Directional derivative of $f = \sin(xy)$ at $(0, \pi)$ in direction $\hat{u} = (1/\sqrt{2}, 1/\sqrt{2})$:

a. π

b. 0

c. $-\pi/\sqrt{2}$

d. $\pi/\sqrt{2}$

32 [Intermediate] • Target: Lines, exponentials, and trig — the toolkit calculus builds on

Net change theorem: $\int_1^4 v(t)dt$ gives the net displacement if $v(t)$ is velocity. For $v = 2t$:

a. 15

b. 8

c. 30

d. 4

33 [Intermediate] • Target: Factoring and rational expressions

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:

a. zero

b. minimum

c. maximum

d. negative

34 [Intermediate] • Target: Inverse functions and logarithms

Carrying capacity K in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$

a. 0

b. K

c. ∞

d. -1

35 [Intermediate] • Target: Piecewise and absolute value review

Rate of change of volume of sphere $V = \frac{4}{3}\pi r^3$ w.r.t r:

a. $4\pi r^2$

b. $\frac{4}{3}\pi r^2$

c. πr^2

d. $8\pi r$

36 [Intermediate] • Target: Systems of equations

Systems: predator-prey (Lotka-Volterra) model has how many equilibria?

a. 2

b. 1

c. 0

d. infinitely many

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37 [Intermediate] • Target: Graphing functions calculus will build on

Find the derivative of $f(x) = \operatorname{arsinh}(x)$ with respect to x.

a. $\frac{1}{\sqrt{x^2+1}}$

b. $\frac{1}{\sqrt{1-x^2}}$

c. $\frac{1}{x^2+1}$

d. $\frac{1}{\sqrt{x^2-1}}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Average rate of change (a preview of the derivative) • Cross-Lesson: 7.01 Integration by Substitution

Related rates: a balloon radius expands at 2 cm/s. Rate of volume change when $r = 5$ cm is:

a. 40π cm³/s

b. 100π cm³/s

c. 200π cm³/s

d. 300π cm³/s

39 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 7.02 Integration by Parts

Frenet equations: $\mathbf{T}' = \kappa\mathbf{N}$, $\mathbf{N}' = -\kappa\mathbf{T} + \tau\mathbf{B}$, $\mathbf{B}' = -\tau\mathbf{N}$. Solving for $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$ given $\kappa(s)$ and $\tau(s)$ determines a curve up to:

a. one translation

b. one rigid motion

c. two parameters

d. rotation only

40 [Challenging] • Target: Interval notation and set operations review • Cross-Lesson: 8.01 Area Between Curves

The Net Change Theorem $\int_a^b F'(x)dx = F(b) - F(a)$ requires F' to be:

a. continuous and positive

b. integrable on $[a, b]$

c. a polynomial

d. bounded and increasing

41 [Challenging] • Target: Functions and composition • Cross-Lesson: 8.02 Volumes of Revolution

Directional derivative $D_{\hat{u}}f = ablaf \cdot \hat{u}$. For $f = x^2y$ at $(1, 1)$ in direction $(1, 1)/\sqrt{2}$:

a. 1

b. $\sqrt{2}$

c. $3\sqrt{2}$

d. $3/\sqrt{2}$

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42 [Challenging] • Target: Lines, exponentials, and trig — the toolkit calculus builds on • Cross-Lesson: 8.03 Arc Length and Improper Integrals

The rate of change in the direction $-\hat{u}$ equals:

a. $D_{\hat{u}}f$

b. 0

c. $|ablaf|$

d. $-D_{\hat{u}}f$

43 [Challenging] • Target: Factoring and rational expressions • Cross-Lesson: 9.01 Separable Differential Equations

Weierstrass substitution $t = \tan(x/2)$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$. Converts rational trig integrands to:

a. trig of t

b. rational functions of t

c. exponentials of t

d. polynomials

44 [Challenging] • Target: Inverse functions and logarithms • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Integration of inverse functions (Laisant-Hermite): $\int_a^b f^{-1}(y)dy = bf^{-1}(b) - af^{-1}(a) - \int_{f^{-1}(a)}^{f^{-1}(b)} f(x)dx$.

Example for $f = e^x$:

a. gives nothing useful

b. yields: $\ln b - \ln a$

c. gives $e^b - e^a$

d. yields: $b\ln b - a\ln a - (b-a)$

45 [Challenging] • Target: Piecewise and absolute value review • Cross-Lesson: 11.01 Sequences and Series

The directional derivative $D_{\hat{u}}f = ablaf \cdot \hat{u}$. Maximum directional derivative equals:

a. $ablaf$

b. $|ablaf|$

c. 1

d. 0

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